

George Galeas

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EDUCATION

California State University, Long Beach |
BS in Aerospace Engineering

Expected Graduation: May 2026

WORK EXPERIENCE

Manufacturing Engineering Intern | Benmo, Long Beach, CA.

September 2023 – February 2024

- Prepared wires; snipping and crimping, for wire harness assemblies.
- Followed detailed schematics to assemble test racks and control panels for various aerospace companies.
- Assembly included the frame build, parts installation, wiring of in-house components, and preparation for delivery.

PROJECTS

Electric UAV for Wildlife Surveillance – Aircraft Design Project |

May 2025

- Designed an electric UAV with the capacity to carry a 35 lbf with a 30 nm range and 20,000ft ceiling.
- Created a python script to size the electric propulsion system, simulate aircraft performance, and verify mission requirements.
- Automated AVL (Athena Vortex Lattice) simulation to find the dynamic wing loading, used to size the airfoil spar.
- Automated airfoil selection with XFOIL and created weight estimations based on Raymer's method to size the aircraft.

FireEye UAV (Hybrid VTOL for Early Wildfire Response) –Senior Design Project |

May 2025

Aerodynamics Lead Engineer

- Created SolidWorks CAD model and conducted airfoil trade study using XFOIL automation.
- Project featured a 100 kg UAV with 7.5-hour endurance, 10 kg payload, and 25 sq. km coverage.

Long Beach Rocketry (LBR) – Competition Club |

June 2024 – Present

Recovery Lead Engineer

- Developed a piston mechanism to eject the nose cone from the airframe, deploying the drogue chute into the airstream.
- Conducted multiple tests including, running electronics in a vacuum chamber, igniting e-matches, and pressurizing our piston.
- Led a team of 5 to design and build a high-powered rocket to test the piston deployment mechanism for our recovery system.

Beach Launch Team (BLT) – Bi-Propellant Liquid Fuel Rocketry Club |

May 2023 – Present

Aerostructures Lead Engineer

- Led a team of 12 to redesign the structural components of the rocket to accommodate for the fluids and avionics systems.
- Conducted a hydrostatic pressure test up to 900 psi on the LOX and fuel tanks to ensure they met our safety factor of 1.5.
- Manage and train subject specific teams for simulation analysis, composite materials research and testing, and manufacturing.

Rocket Main Structure Manufacturing

- Programmed a CNC router to cut out 7 wooden bulkheads for the structural support of the assembly.
- Worked in a team of 4 to reinforce the propellant tanks with fiberglass, using a prepreg carbon fiber.
- Fabricated fiberglass fins and attached them onto the airframe using a tip-to-tip layup process.

FEA and CFD of Rocket

- Used Ansys Mechanical to test individual components and the full rocket assembly for maximum stress and deformation.
- Used Ansys Fluent for CFD simulation to find the drag coefficient and moment of the rocket to calculate the required thrust.

SKILLS AND CERTIFICATIONS

Certifications: SOLIDWORKS CAD Design Professional (CSWP), SOLIDWORKS CAD Design Associate (CSWA)

Software: SolidWorks, Autodesk Inventor, Fusion 360, OpenRocket, Ansys, Python, MATLAB, XFOIL, AVL, Microsoft Office

Manufacturing: Manual Mill, Manual Lathe, CAM, CNC Router, 3D Printing, Fiberglass Layups, Vacuum Bagging